**Table S1.** Characteristics of patients whose tumors were used for isolation of CAFs and RNA-seq

| Patients ID | Age | RNA-seq | Tumor size (cm) | ER | PR | HER2 | Lymph node metastasis | Histology grade |
|---|---|---|---|---|---|---|---|---|
| BC01 | 48 | Yes | 4.3*1.8*1.3 | + | + | - | - | III |
| BC02 | 41 | Yes | 2.1*2.0*1 | - | - | - | - | II |
| BC03 | 67 | Yes | 2.2*1.6*1.2 | + | + | - | - | III |
| BC04 | 74 | No | 2.3*2*1.4 | + | - | - | - | III |
| BC05 | 57 | Yes | 2.1*1.8*1.2 | + | + | - | + | II |
| BC06 | 44 | Yes | 2.4*1.5*1 | + | + | + | + | III |
| BC07 | 48 | Yes | 2.5*2*2 | - | - | + | + | III |
| BC08 | 56 | No | 2*1.5*1 | + | + | - | + | II |

**Table S2.** Characteristics of patients whose tumors were used for construction of the tissue microarray

| Patients number | Gender | Age | Neoadjuvant chemotherapy | ER (%) | PR (%) | HER2 | Ki-67 (Li) | Lymph node metastasis | TNM Stage |
|---|---|---|---|---|---|---|---|---|---|
| P01 | female | 56 | - | + | - | + | 30% | + | T2N1M0 |
| P02 | female | 58 | - | - | + | + | 20% | - | T1N0M0 |
| P03 | female | 51 | - | + | - | - | 15% | - | T2N0M0 |
| P04 | female | 55 | - | + | + | + | 35% | - | T1N0M0 |
| P05 | female | 58 | - | - | - | - | 70% | + | T1N1M0 |
| P06 | female | 40 | + | + | - | - | 90% | + | T2N1M0 |
| P07 | female | 45 | - | + | + | - | 30% | - | T1N0M0 |
| P08 | female | 57 | - | + | - | + | 20% | - | T1N0M0 |
| P09 | female | 72 | + | + | + | - | 5% | + | T2N2M0 |
| P10 | female | 59 | - | + | + | - | 10% | - | T2N0M0 |
| P11 | female | 68 | - | + | + | - | 15% | + | T1N2M0 |
| P12 | female | 48 | - | + | + | - | 30% | - | T2N0M0 |
| P13 | female | 30 | - | - | - | + | 15% | - | T1N0M0 |
| P14 | female | 65 | - | + | + | + | 30% | - | T2N0M0 |
| P15 | female | 46 | - | + | + | - | <5% | - | T1N0M0 |
| P16 | female | 57 | - | + | + | + | 80% | + | T2N1M0 |
| P17 | female | 68 | - | + | + | - | 40% | + | T1N1M0 |
| P18 | female | 65 | - | NA | NA | NA | - | - | Tis |
| P19 | female | 38 | + | + | + | - | 20% | - | T2N0M0 |
| P20 | female | 50 | - | - | - | - | 60% | - | T2N0M0 |
| P21 | female | 66 | - | - | - | + | 20% | + | T2N1M0 |
| P22 | female | 49 | - | + | + | - | 5% | - | T1N0M0 |
| P23 | female | 47 | - | + | - | + | 40% | + | T2N2M0 |
| P24 | female | 56 | - | + | + | + | 25% | - | T2N0M0 |
| P25 | female | 69 | - | + | + | - | 30% | - | T1N0M0 |
| P26 | female | 43 | - | + | + | - | 40% | + | T1N1M0 |
| P27 | female | 49 | - | NA | NA | NA | - | - | Tis |
| P28 | female | 28 | + | + | + | - | 20% | - | T1N0M0 |
| P29 | female | 57 | + | + | + | - | 30% | - | T1N0M0 |
| P30 | female | 49 | - | - | - | + | 20% | + | T2N1M0 |
| P31 | female | 65 | - | + | + | - | 5% | - | T2N0M0 |
| P32 | female | 62 | - | + | + | - | 20% | - | T1N0M0 |
| P33 | female | 32 | + | - | - | + | 40% | + | T3N1M0 |
| P34 | female | 51 | - | + | + | - | 60% | - | T1N0M0 |
| P35 | female | 50 | - | + | + | - | 15% | + | T1N2M0 |

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| P36 | female | 51 | - | + | + | - | 10% | - | T2N0M0 |
| P37 | female | 64 | - | + | + | - | 20% | + | T1N1M0 |
| P38 | female | 78 | - | + | + | - | 10% | + | T2N2M0 |
| P39 | female | 45 | - | + | + | - | 10% | + | T1N2M0 |
| P40 | female | 43 | - | + | + | + | >10% | + | T2N2M0 |
| P41 | female | 62 | - | + | + | - | 10% | - | T2N0M0 |
| P42 | female | 55 | - | + | + | - | 25% | - | T1N0M0 |
| P43 | female | 47 | - | + | + | - | >15% | + | T2N1M0 |
| P44 | female | 40 | - | + | + | - | <10% | + | T2N2M0 |
| P45 | female | 83 | - | + | - | - | 30% | - | T2N0M0 |
| P46 | female | 45 | - | + | - | + | 50% | + | T1N1M0 |
| P47 | female | 48 | - | + | - | - | 20% | + | T2N2M0 |

Blue areas are patients with neoadjuvant chemotherapy.

**Table S3.** Correlation between MXRA5 expression and clinicopathologic characteristics of breast cancer patients in TCGA

| Characteristics | Low expression of MXRA5 | High expression of MXRA5 | *P* value |
|---|---|---|---|
| n | 543 | 544 | |
| Pathologic T stage, n (%) | | | 0.008 |
| T1 | 118 (10.9%) | 160 (14.8%) | |
| T2 | 323 (29.8%) | 308 (28.4%) | |
| T3&T4 | 99 (9.1%) | 76 (7%) | |
| Pathologic N stage, n (%) | | | 0.005 |
| N0 | 272 (25.5%) | 244 (22.8%) | |
| N1 | 183 (17.1%) | 176 (16.5%) | |
| N2 | 40 (3.7%) | 76 (7.1%) | |
| N3 | 38 (3.6%) | 39 (3.7%) | |
| Pathologic M stage, n (%) | | | 0.887 |
| M0 | 438 (47.4%) | 467 (50.5%) | |
| M1 | 10 (1.1%) | 10 (1.1%) | |
| Pathologic stage, n (%) | | | 0.287 |
| Stage I | 85 (8%) | 97 (9.1%) | |
| Stage II | 327 (30.8%) | 292 (27.5%) | |
| Stage III | 114 (10.7%) | 130 (12.2%) | |
| Stage IV | 9 (0.8%) | 9 (0.8%) | |
| Age, n (%) | | | 0.063 |
| ≤ 60 | 286 (26.3%) | 317 (29.2%) | |
| > 60 | 257 (23.6%) | 227 (20.9%) | |
| HER2 status, n (%) | | | 0.050 |
| Negative | 263 (36.1%) | 297 (40.7%) | |
| Indeterminate | 8 (1.1%) | 4 (0.5%) | |
| Positive | 60 (8.2%) | 97 (13.3%) | |

**Table S4.** Univariate analysis and multivariate analysis of the correlation of MXRA5 expression with OS among breast cancer

| Characteristics | Total (N) | Univariate analysis | | Multivariate analysis | |
|---|---|---|---|---|---|
| | | Hazard ratio (95% CI) | *P* value | Hazard ratio (95% CI) | *P* value |
| Pathologic T stage | 1,083 | | | | |
| T1 | 277 | Reference | | Reference | |
| T2 | 631 | 1.334 (0.889-2.003) | 0.164 | 0.942 (0.519-1.710) | 0.844 |
| T3&T4 | 175 | 1.931 (1.208-3.088) | 0.006 | 2.111 (1.023-4.355) | 0.043 |
| Pathologic N stage | 1,067 | | | | |
| N0 | 516 | Reference | | Reference | |
| N1 | 358 | 1.947 (1.322-2.865) | < 0.001 | 1.613 (0.927-2.809) | 0.091 |
| N2 | 116 | 2.522 (1.484-4.287) | < 0.001 | 1.837 (0.849-3.978) | 0.123 |
| N3 | 77 | 4.191 (2.318-7.580) | < 0.001 | 4.115 (1.694-9.995) | 0.002 |
| Pathologic M stage | 925 | | | | |
| M0 | 905 | Reference | | Reference | |
| M1 | 20 | 4.266 (2.474-7.354) | < 0.001 | 3.389 (1.338-8.588) | 0.010 |
| Age | 1,086 | | | | |
| ≤ 60 | 603 | Reference | | Reference | |
| > 60 | 483 | 2.024 (1.468-2.790) | < 0.001 | 2.838 (1.735-4.643) | < 0.001 |
| HER2 status | 717 | | | | |
| Negative | 560 | Reference | | Reference | |
| Positive | 157 | 1.593 (0.973-2.609) | 0.064 | 1.254 (0.716-2.197) | 0.428 |
| MXRA5 | 1,086 | | | | |
| Low | 542 | Reference | | Reference | |
| High | 544 | 1.340 (0.971-1.848) | 0.075 | 1.083 (0.659-1.780) | 0.753 |

**Table S5.** Clinical information for breast cancer patients were used for RT-qPCR

| Patients ID | Age | Neoadjuvant chemotherapy | Tumor size (cm) | ER (%) | PR (%) | HER2 | Ki-67 (Li) | Lymph node metastasis | Histology grade |
|---|---|---|---|---|---|---|---|---|---|
| 01 | 38 | No | 3.5*2.5*2.5 | - | - | 3+ | 30% | + | III |
| 02 | 67 | No | 3.5*2.2*1.5 | - | - | - | 30% | - | III |
| 03 | 35 | No | 3.5*3*2 | + | + | 2+, FISH- | 10% | - | II |
| 04 | 61 | No | 3*2*1.5 | - | - | 2+, FISH- | 30% | - | III |
| 05 | 71 | No | 2*2*1.5 | + | + | 1+ | 30% | - | II |
| 06 | 67 | No | 3*2.8*2.5 | + | + | 2+, FISH- | 30% | + | II |
| 07 | 51 | No | 3*1.5*1.5 | + | + | 3+ | 70% | + | III |
| 08 | 63 | No | 3.5*2.7*2 | + | - | - | 30% | + | III |
| 09 | 31 | No | 1.5*1.5*1 | + | + | 2+, FISH+ | 30% | + | II |
| 10 | 71 | No | 2.5*1.5*1 | + | - | 2+, FISH- | 15% | + | II |
| 11 | 38 | No | 2.2*2*1.5 | + | + | - | 20% | - | II |
| 12 | 52 | No | 3.3*3*2.5 | + | + | 2+, FISH+ | 20% | - | III |
| 13 | 71 | No | 2.5*1.7*1 | + | + | 1+ | 10% | + | II |
| 14 | 40 | No | 2.5*1.5 | + | + | 1+ | <10% | + | II |
| 15 | 51 | No | 2.5*2*1.5 | + | + | - | 10% | - | II |

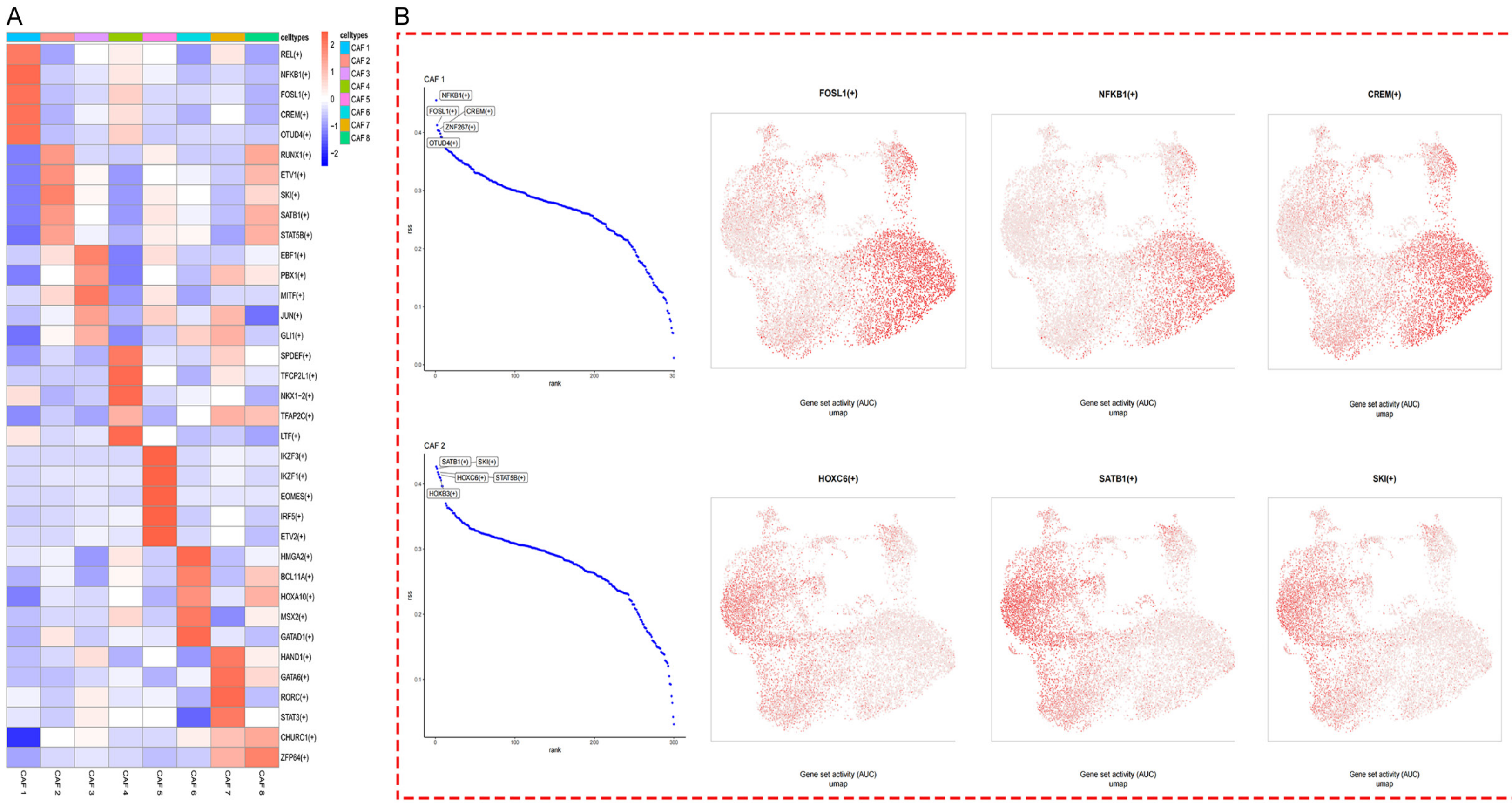
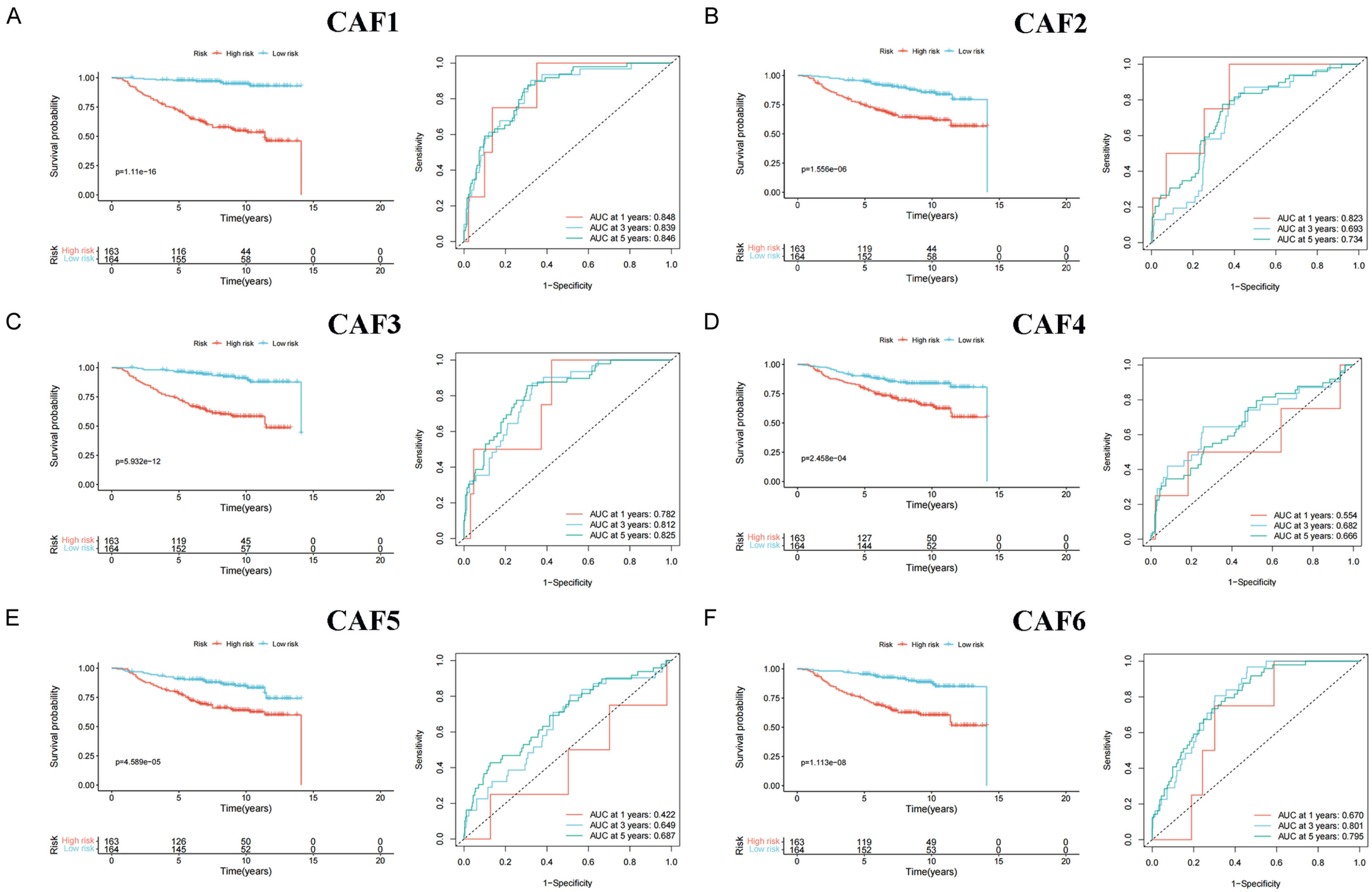



**Figure S1.** Transcription factors associated with caf isoforms in breast cancer. A. Heatmap of gene expression of transcription factors associated with CAF1-8 cell subtypes. B. Transcription factors such as NFKB1 and SATB1 were significantly expressed in CAF1-2.


A
CAF1
Risk High risk Low risk
Survival probability
p=1.11e-16
Time(years)
High risk 163 116 44 0 0
Low risk 164 155 58 0 0
Sensitivity
1−Specificity
AUC at 1 years: 0.848
AUC at 3 years: 0.839
AUC at 5 years: 0.846
B
CAF2
p=1.556e-06
High risk 163 119 44 0 0
Low risk 164 152 58 0 0
AUC at 1 years: 0.823
AUC at 3 years: 0.693
AUC at 5 years: 0.734
C
CAF3
p=5.932e-12
High risk 163 119 45 0 0
Low risk 164 152 57 0 0
AUC at 1 years: 0.782
AUC at 3 years: 0.812
AUC at 5 years: 0.825
D
CAF4
p=2.458e-04
High risk 163 127 50 0 0
Low risk 164 144 52 0 0
AUC at 1 years: 0.554
AUC at 3 years: 0.682
AUC at 5 years: 0.666
E
CAF5
p=4.589e-05
High risk 163 126 50 0 0
Low risk 164 145 52 0 0
AUC at 1 years: 0.422
AUC at 3 years: 0.649
AUC at 5 years: 0.687
F
CAF6
p=1.113e-08
High risk 163 119 49 0 0
Low risk 164 152 53 0 0
AUC at 1 years: 0.670
AUC at 3 years: 0.801
AUC at 5 years: 0.795

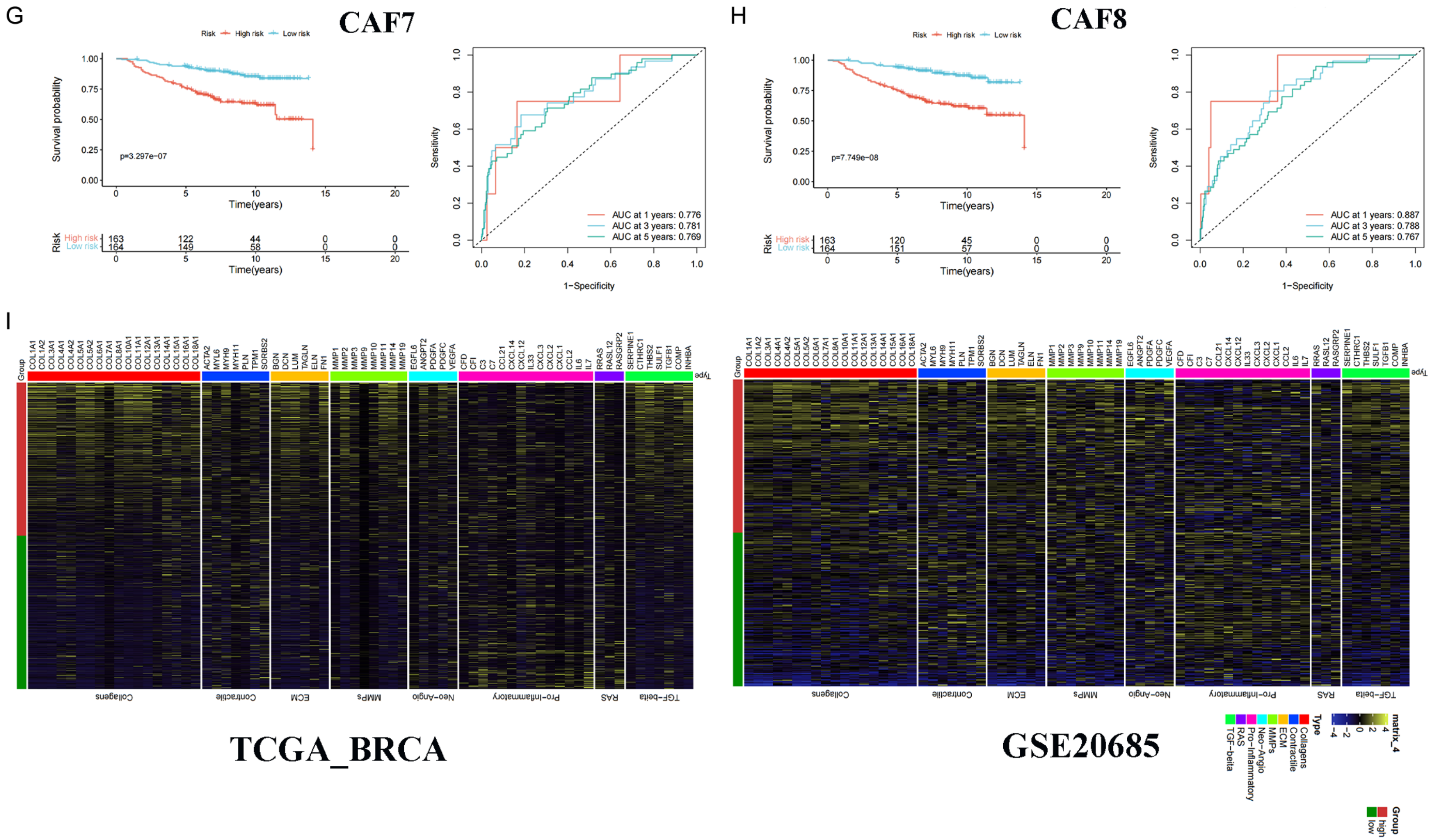



**Figure S2.** Expression of cancer-promoting genes in breast cancer with high CAF2 scores. A-H. Survival curves for the GES20685 dataset showed that among breast cancers with different CAF subtypes, the group subtype with a higher score had a worse prognosis. I. Heatmap of gene expression associated with collagen, ECM, MMPs, TGF-β, Neo-Angio, Contractile, RAS, and Pro-Inflammatory certain functions in the TCGA-BRCA and GSE20685 datasets in the high-CAF2 scoring subgroups.

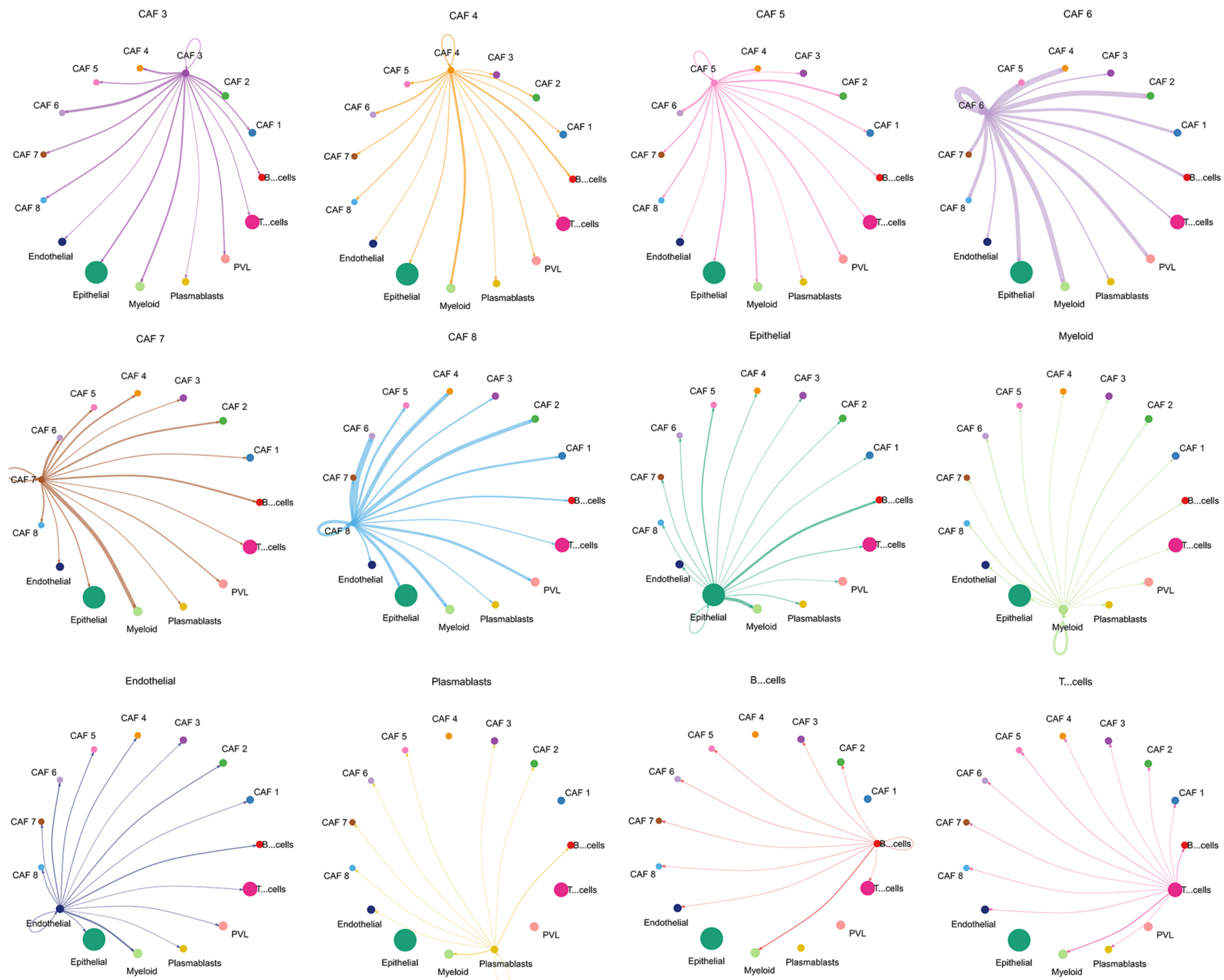



**Figure S3.** The strength of interactions between individual subclass cell populations and other subclass cell populations.